



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

Fax: +66-2721-3515

SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Product name: fluazifop-P-butyl 15% W/V EC

Product use: Herbicide

Relevant identified uses of the substance/mixture and uses advised against:

Used as a selective post-emergence for control of annual and perennial grasses. May be applied over the top of many ornamentals including many woody and herbaceous ornamentals in container and field nurseries, landscapes and greenhouses. May also be used for bermudagrass suppression in tall fescue turf (see label for specific guidelines).

2. HAZARD IDENTIFICATION

Hazard classification:

Hazardous. Harmful, dangerous for the environment.

Risk Phrases:

R22: Harmful if swallowed.

R36: Irritating to eyes.

R43: May cause sensitization by skin contact.

R50/53: Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

R63: Possible risk of harm to the unborn child.

Safety Phrases:

S2: Keep out of reach of children

S13: Keep away from food, drink and animal feeding stuff

S20/21: When using do not eat, drink or smoke.

S26: In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

S35: This material and its container must be disposed of in a safe way.



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S61: Avoid release to the environment. Refer to special instructions/safety data sheets.

3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: (*R*)-2-[4-(trifluoromethyl-2-pyridyloxy)phenoxy]propionate [IUPAC]

Common name: fluazifop-P-butyl

CAS Number: 83066-88-0

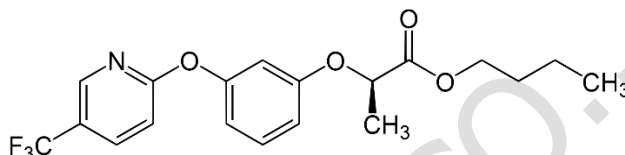
Ingredients and composition: Active ingredient - fluazifop-P-butyl 150 g/L

Chemical family: Aryloxyphenoxypropionate

Formula: $C_{19}H_{20}F_3NO_4$

Molecular weight: 383.367 g/mol

Structural formula:



4. FIRST AID MEASURES

Description of first aid measures

In case of swallowed:

If swallowed, call a Poison Control Center or doctor immediately for treatment advice. Have the person sip a glass of water if able to swallow. DO NOT induce vomiting unless told to do so by a Poison Control Center or doctor.

In case of eyes contact:

If in eyes, hold eye open and rinse slowly and gently with water for 15 – 20 minutes. Remove contact lenses, if present, after 5 minutes, then continue rinsing eye. Call a Poison Control Center or doctor for treatment advice.

In case of skin contact:

If on skin or clothing, take off contaminated clothing. Rinse skin immediately with plenty of water for 15 – 20 minutes. Call a Poison Control Center or doctor for treatment advice.

In case of Inhaled contact:

If inhaled, move person to fresh air. If person is not breathing, call an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a Poison Control Center for further treatment advice.

Note to Physician:

There is no specific antidote. Treat symptomatically.



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5. FIRE FIGHTING MEASURES

Flammability Limits: Not available

Auto-ignition temperature: Not available

Fire point: > 61 °C

Unusual Fire, Explosion and Reactivity Hazards:

During a fire, irritating and possibility toxic gases may be generated by thermal decomposition or combustion.

In Case of Fire:

Use dry chemical, foam or CO₂ extinguishing media. Wear full protective clothing and self-contained breathing apparatus. Evacuate non-essential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated building, area, and equipment until decontaminated. Water runoff can cause environmental damage. If water is used to fight fire, dike and collect runoff.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Emergency Procedures:

In case of spillage it is important to take all steps necessary to:

- Avoid eye and skin contact.
- Avoid contamination of waterways.

Methods and Materials for Containment and Clean Up:

Procedure for spill:

- (1) Keep all bystanders away.
- (2) Wear goggles, half face-piece respirator with combined dust and vapor cartridge, full length clothing and PVC gloves.
- (3) Re-position any leaking containers so as to minimize further leakage.
- (4) Dam and absorb spill with an absorbent material (e.g. sand or soil).
- (5) Shovel the absorbed spill into drums.
- (6) Disposal of the absorbed material will depend upon the extent of the spill.
 - For quantities up to 50L of product bury in a secure landfill site.
 - For quantities greater than 50L seek advice from the manufacturer (use emergency contact number below) before attempting disposal. Contain in a secure location until disposal method is established.
- (7) Decontaminate the spill area with detergent and water and rinse with water.



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7. HANDLING AND STORAGE

Store the material in a well-ventilated, secure area out of reach of children and domestic animals. DO NOT store food, beverages or tobacco products in the storage area. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Exposure standards

No exposure standards have been set for fluazifop-P-butyl technical.

Engineering controls

Provide adequate ventilation to keep airborne concentration below the exposure limit.

Personal Protective Equipment

Will irritate the eyes and skin. Avoid contact with eyes and skin. If product in eyes, wash it out immediately with water. When opening the container and preparing spray, wear:

- Cotton overalls buttoned to the neck and wrist and a washable hat
- Elbow-length PVC gloves and goggles
- Self-contained breathing apparatus and total protection required in enclosed area

When using (except aerial spraying) the prepared spray, wear:

- Cotton overalls buttoned to the neck and wrist and a washable hat and elbow-length PVC gloves.
- Wash hands after use. After each day's use, wash gloves, goggles and contaminated clothing.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical stage:	Liquid
Color:	Brown
Odor:	Hydrocarbon solvent
pH:	5.0 to 8.0
Boiling point:	154 °C at 0.02 mmHg
Melting/ Freezing Point:	-20 °C (glass-like)
Density:	0.95 (20 °C)
Vapor pressure:	0.033 mPa at 20 °C
Solubility in water:	Emulsifies in water



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10. STABILITY AND REACTIVITY

Stability:	Stable under normal use and storage.
Hazardous Polymerization:	Will not occur.
Conditions to avoid:	Avoid high temperature.
Materials to avoid:	Avoid strong oxidizer and alkaline.
Hazardous Decomposition Products:	Will evolve toxic and irritant vapors.

11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, fluazifop-P-butyl:

LD50 (Oral, rat):	> 2,000 mg/kg
LD50 (Dermal, rat):	> 2,000 mg/kg
LC50 (Inhalation, rat, 4h):	> 5.31 mg/L
Skin irritation (rabbit):	Not irritant.
Eye irritation (rabbit):	Not irritant.
Skin sensitization:	Skin sensitizer to Guinea pig

12. ECOLOGICAL INFORMATION

Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to fluazifop-P-butyl technical:

Fish – Acute toxicity:

Rainbow trout,

LC50 (96 h): 1.3 mg/L

Daphnia magna,

EC50 (48 h): > 1.0 mg/L

Birds – Acute toxicity:

Mallard ducks,

LD50: > 3,500 mg/kg

Algae – Acute toxicity:

Navicula pelliculosa,

EbC50 (72 h): 0.51 mg/L

Earthworm – Acute toxicity:

LC50: > 1,000 mg/kg



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Honey bee – Acute toxicity:

Very low toxic to bees

LD50: > 0.2 mg/bee (oral and contact)

Other aquatic spp. – Acute toxicity:

Lemna gibba,

EC50 (14 days): > 1.4 mg/L

Soil/Environment:

Fluazifop-P-butyl, is relatively non-persistence in soil; reported half-life < 1 week. Fluazifop-P-butyl breaks down or hydrolyzes rapidly in soils or water to the fluazifop acid, which is also low persistence. Fluazifop-P-butyl and fluazifop-P have low mobility in soils and not to present appreciable risks for groundwater contamination.

13. DISPOSAL CONSIDERATIONS

Dispose of empty, used containers by:

Single rinse before disposal. Add rinsings to spray tank. DO NOT dispose of undiluted chemicals on site. Puncture or shred and bury empty containers in a local authority landfill. If no landfill is available, bury the containers below 500 mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt.

14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

Land transport ADR/RID

UN No:	3082
Shipping Name:	Environmentally Hazard Substance, Liquid, N.O.S. (contains fluazifop-P-butyl)
Dangerous Goods Class:	9
Subsidiary Risk:	Not regulated
Packaging Group:	III

Marine transport IMDG

UN No:	3082
Shipping Name:	Environmentally Hazard Substance, Liquid, N.O.S. (contains fluazifop-P-butyl)
Dangerous Goods Class:	9



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Subsidiary Risk: Not regulated

Packaging Group: III

Marine pollutant: Yes

Air transport ICAO/IATA

UN No: 3082

Shipping Name: Environmentally Hazard Substance, Liquid, N.O.S. (contains fluazifop-P-butyl)

Dangerous Goods Class: 9

Subsidiary Risk: Not regulated

Packaging Group: III

15. REGULATORY INFORMATION

EPCRA SARA Title III Classification

Section 311/312 Hazard Classes: Acute Health Hazard
Chronic Health Hazard

Section 313 Toxic Chemicals: Not applicable

California Proposition 65

Not applicable

CERCLA/SARA 302 Report Quantity (RQ)

None

TSCA Status

Exempt from TSCA, subject to FIFRA

16. OTHER INFORMATION

This information is provided in good faith but without express or implied warranty. Buyer assumes all responsibility for safety and use not in accordance with label instructions.